

Jaisalmer Oil & Gas Risk & Status Overview

Heavy crude, desert logistics, failed exploration, and current production

Project Report

Purpose: A structured presentation of the supplied information on oil and gas operations and risk profiles in Jaisalmer district, Rajasthan.

1. Red (High Risk: Heavy Crude Plugging)

The Cases: Extraction operations in the **Baghewala field**.

The Details: The Baghewala location contains highly viscous, heavy oil (similar to bitumen). It requires intense thermal stimulation (like injecting steam) to make the oil thin enough to pump. If these heating systems fail, the heavy crude rapidly cools and solidifies inside the pipes, completely plugging the wellbore and risking total mechanical failure.

2. Yellow (Mid Risk: Severe Desert Logistics)

The Cases: Drilling rigs and surface facilities across the **Tanot and Dandewala fields**.

The Details: Operating deep within the Thar Desert in the Jaisalmer district introduces immense environmental strain. Operators battle extreme summer heat (regularly exceeding 50°C) which stresses machinery, severe sandstorms that degrade exposed rig equipment, and acute water scarcity that makes mixing drilling mud very difficult.

3. Brown (Abandoned: Failed Exploration)

The Cases: Legacy exploratory dry holes drilled by Oil India Limited (OIL) along the outer flanks of the **Jaisalmer Basin**.

The Details: Geologists drilled into promising limestone and sandstone formations based on historical seismic data. However, these specific exploratory wells were permanently plugged and abandoned because they hit rock formations that were completely dry, or they found heavy oil reserves that were too thick and uncommercial to pump to the surface.

4. Green (Active & Successful: Current Production)

The Cases: Actively producing natural gas wells in the **Tanot and Dandewala fields**, alongside specialized heavy-oil extraction wells in **Baghewala**.

The Details: This region is a major onshore success story. The high-quality sweet natural gas continuously pumped from the Tanot and Dandewala fields is pipelined directly to the Ramgarh Gas Thermal Power Station, acting as a critical fuel source for the local electricity grid.

5. Risk & Status Summary

Category	Location / Cases	Main Issue or Status
Red — High Risk	Baghewala field	Heavy crude requires thermal stimulation; cooling can cause plugging.
Yellow — Mid Risk	Tanot & Dandewala fields	Extreme heat, sandstorms, and water scarcity create logistics and equipment challenges.
Brown — Abandoned	Outer flanks of Jaisalmer Basin	Exploratory dry holes and heavy oil considered too thick/uncommercial.
Green — Active & Successful	Tanot, Dandewala & Baghewala	Natural gas production and specialized heavy-oil extraction are active.

6. Frontend / Database Mapping Note

To map these specific Jaisalmer district well outcomes and risk profiles to your project's frontend or database architecture, reference your file named "What actually i have to do for my web".

Content note: This PDF preserves the information supplied in the request. The technical and operational claims have not been independently verified during this conversion.